

G Sreedev

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SUMMARY

Computer Science and Engineering graduate specializing in AI & Machine Learning, with hands-on experience in Full Stack Applications, backend development. I like building software beyond just the interface, understanding the APIs, databases, and deployment that make a system work as a whole. My experience includes working with Python, FastAPI, Django, React, PostgreSQL, Docker across professional and academic projects. I'm looking to apply this practical experience to building reliable software and solving meaningful engineering problems.

TECHNICAL SKILLS

Languages: Python, JavaScript
Backend Technologies: FastAPI, Django, JWT
Frontend: React.JS, CSS
Databases: PostgreSQL
Tools: Git, GitHub, Docker

EXPERIENCE

Software Engineer Intern <i>QuanHack Solutions</i>	May 2026 – Present <i>Kochi, Kerala</i>
- Fixed a concurrency bug in a multi-worker orchestrator using atomic MongoDB updates, preventing overwritten step statuses and duplicate execution. - Implemented prompt injection protection for an LLM-driven dynamic capability system by adding recursive pattern-based scanning of generated and user-edited definitions, blocking malicious payloads before they reach the database. - Improved LLM request reliability by replacing custom retry loops with Tenacity-based exponential backoff (up to 5 attempts, 60s max delay), while preserving OpenRouter-to-Gemini fallback behavior.	

PROJECTS

IoT Based Real time Air Pollution Monitoring System with Machine Learning	Dec 2025 – May 2026
- Designed and implemented a real-time pollution monitoring system integrating Arduino-based gas sensors (MQ135, MQ7, DHT22) with a Flask backend and Flutter mobile application. - Applied Random Forest classification for air quality prediction with 92.14% accuracy. - Developed live AQI visualization, location-based monitoring, and alert features for users.	
AI Driven Disease Prediction and Doctor Consultancy System	Nov 2025
- Built a Django-based healthcare web app with server-side rendering that predicts diseases from user-reported symptoms, handling both the UI and prediction logic within a single monolithic architecture. - Integrated gemini-1.5-flash to generate accurate, real-time health insights. - Implemented a doctor consultancy and appointment booking system with secure patient data management.	
E-Commerce Web Application	Nov 2025
- Developed a Django-based E-commerce web application with responsive UI using Bootstrap. - Implemented product listing with search, pagination, and dynamic cart features (add, increment, decrement, remove). - Built a custom admin panel with full CRUD operations for product management. - Used SQLite and Django ORM for efficient data handling and session-based cart management.	

PUBLICATIONS

A Hybrid IoT and Machine Learning Approach for Real-Time Air Pollution Detection and Prediction

- Co-authored an IEEE conference paper proposing a low-cost IoT and ML-based air pollution monitoring system using Arduino sensors, K-Means clustering, Random Forest regression, and a Flutter mobile application for real-time monitoring and prediction, achieving an R^2 score of 0.92.

EDUCATION

B.Tech. in Computer Science and Engineering (AI & ML)

Sree Buddha College of Engineering, Pattoor

Sept. 2022 – May 2026

CGPA: 7.4/10

Higher Secondary Education

Sree Buddha Central School, CBSE

May 2020 – Feb 2022

Percentage: 75

High School

Sree Buddha Central School, CBSE

April 2019 – Feb 2020

Percentage: 94

CERTIFICATIONS

- **Full Stack Web Development with AI Tools** – Next Gen Employability Program
- **Python for Data Science, AI & Development** – IBM, Coursera
- **The Complete Python Bootcamp: From Zero to Hero in Python** – Udemy
- **APIs** – Meta, Coursera